

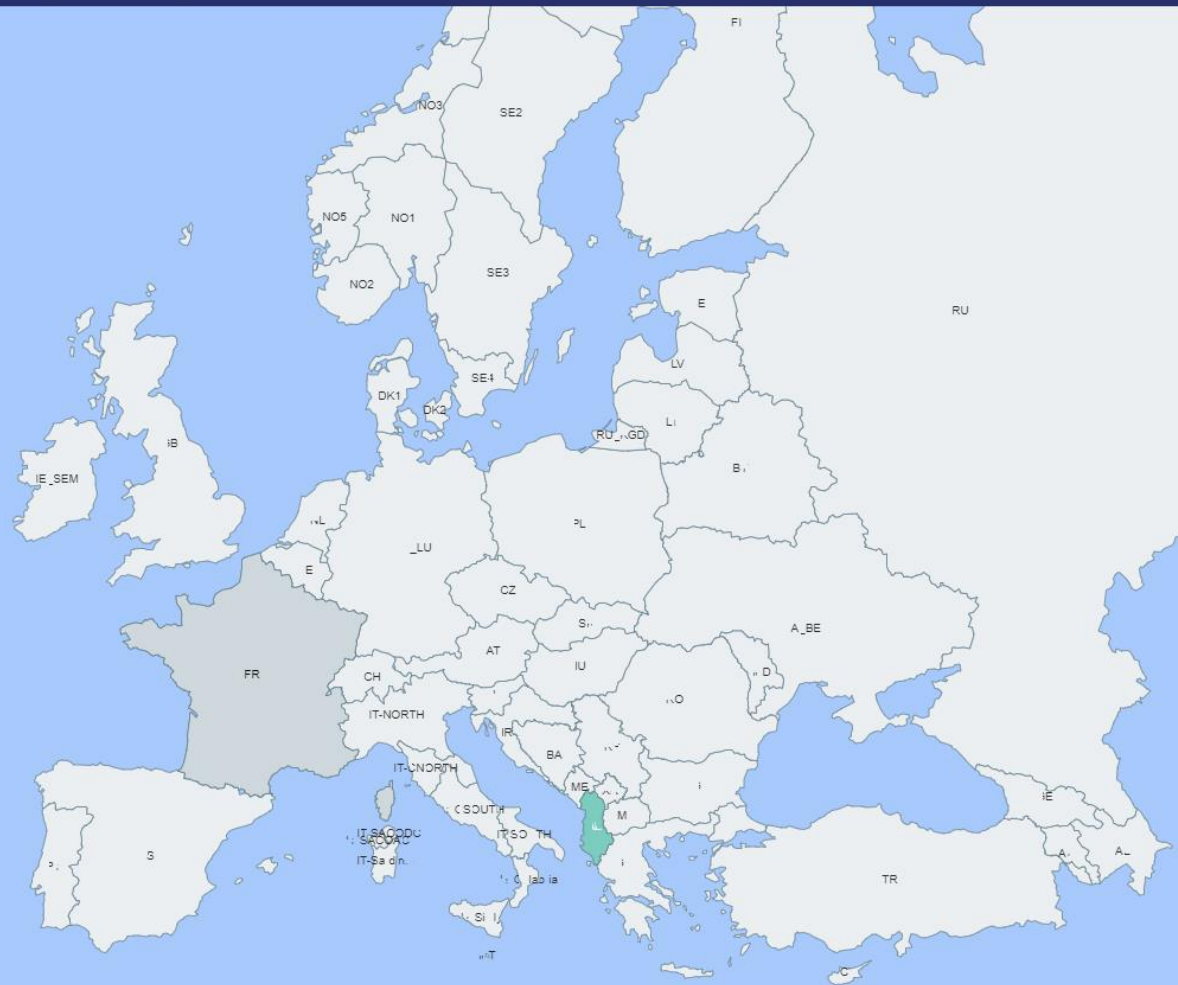


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Semester Project 24/25

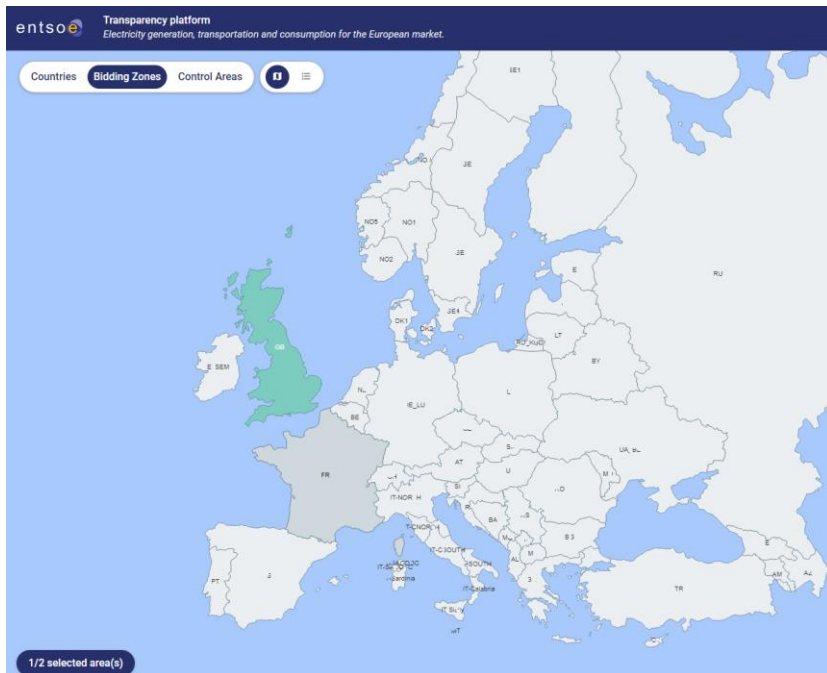
Energy Data Engineering 1 | Prof. Dr. Manuel Lämmle

Bidding Zones



Select multiple areas

Project description

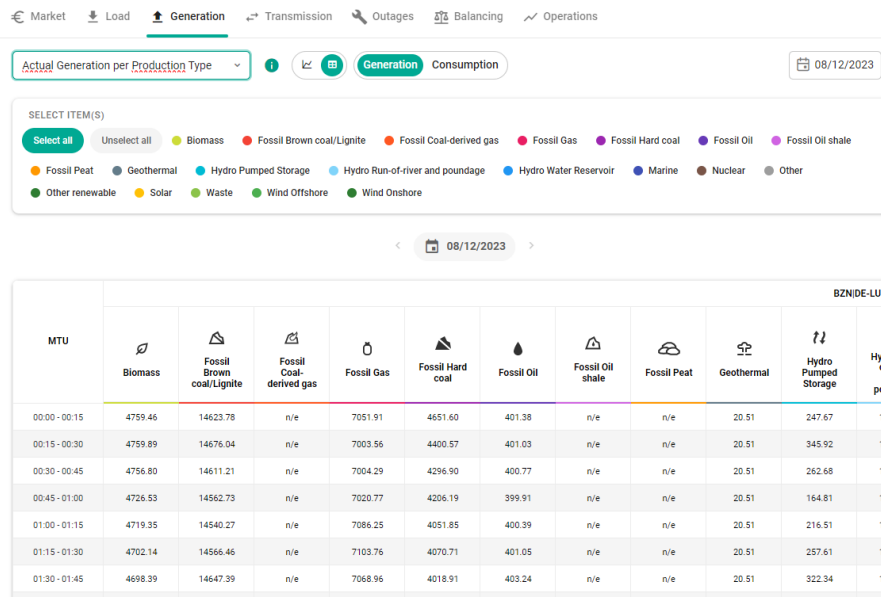


10 Countries/Bidding zones:

1. BE - Belgium
2. CZ – Czech Republic
3. DE-LU - Germany and Luxembourg
4. DK1 – Denmark 1
5. ES - Spain
6. FR – France
7. FI - Finland
8. GR - Greece
9. PL - Poland
10. PT – Portugal
11. AT – Austria
12. HU - Hungary

➔ 1 Country per Team

Project data



► Source of data:
<https://newtransparency.entsoe.eu/>

► Relevant data sets:

► **Actual Generation per Production Type:**
<https://transparency.entsoe.eu/generation/r2/actualGenerationPerProductionType/show>

► **Day-ahead Prices:**
<https://transparency.entsoe.eu/load-domain/r2/totalLoadR2/show>

► **Total load - Day-ahead/Actual:**
<https://transparency.entsoe.eu/transmission-domain/r2/dayAheadPrices/show>

➔ Download hourly data from ENTSOE

Project goal



- ▶ Carry out a data engineering project in a team of 3 students.
- ▶ Your goal is to analyze the electricity system in Europe applying data analysis and data engineering methods
- ▶ Organize the data and your team independently
- ▶ Work on the four tasks on the following slides

Project deliverables



Files to upload:

- ▶ Jupyter Notebook
 - ▶ Functional Code (also needs to run on my computer!)
 - ▶ Documentation using markup (see evaluation criteria)
 - ▶ Source files
- ▶ Html version of notebook

▶ Project presentation

- ▶ Maximum 20 mins per team.
- ▶ All team members have to present
- ▶ Focus on results and methods, not the code
- ▶ You don't have to present everything, but focus on highlights and relevance
- ▶ No extra slides, use your Notebook only

Due date: 21. January (submission)

Presentation: 22. January

Oral: 22. January (afternoon)

Evaluation Criteria



Submitted Notebooks

- ✓ Correctness
- ✓ Completeness
- ✓ Conciseness
- ✓ Creativity of solutions
- ✓ Style of Programming
- ✓ Documentation of code
- ✓ Application of theory and methods
- ✓ Documentation in Appendix:
 - ✓ Involvement per Task:
 - ✓ Student 1: x %, Student 2: y %
 - ✓ Usage of ChatGPT:
 - ✓ ChatGPT is allowed, but...
 - ✓ Usage must be made transparent: when and how ChatGPT was included?
- ✓ Sources

- ▶ Individual grade for presentation
- ▶ Individual grade for oral examination

Task 1:

Exploratory data analysis of energy generation for the year 2024

1. Calculate the share of renewables
2. Plot the time-series for energy generation showing all energy carriers
3. Plot the electricity mix (i.e. share per energy carrier)
4. Conduct a exploratory data analysis on:
 1. Day-Ahead Price
 2. Share of renewables
5. Analyze the hypothesis, that the day-ahead price correlates with the share of renewables by using appropriate statistical methods

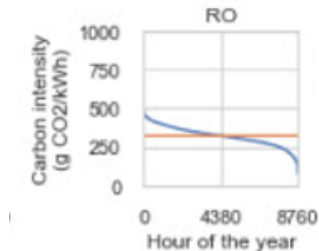
Task 2: CO2 intensity for electricity generation in 2024



- ▶ Calculate the dynamic CO2 intensity of generated electricity in g/kWh for every hourly time step:

$$co2(t) = \frac{\sum_{technology} co2_{intensity_factor, technology} \cdot P_{technology}(t)}{\sum_{technology} P_{technology}(t)}$$

- ▶ Use the CO2 emission factors provided in the table on the next slide.
- ▶ Analyze the CO2 intensity of electricity mix using adequate visualization and statistics
 - ▶ Average grid intensity of the country/bidding zone?
 - ▶ Plot a CO2 intensity duration curve (compare picture, Wikipedia: “Load duration Curve”)



CO2 emission factors



Source:

<https://www.sciencedirect.com/science/article/pii/S2772783122000176>

Country	Technology	IP (GtCO ₂ /year)	Coverage (%)	Flexibility	Dispatch	Dispatch	Dispatch	Dispatch	Dispatch
Country	Technology	IP (GtCO ₂ /year)	Coverage (%)	Flexibility	Dispatch	Dispatch	Dispatch	Dispatch	Dispatch
AT	gas	288.29	24.0	1.954	7	3.903	17	3.903	17
AT	hard_coal	288.27	24.0	1.9	1	0.003	1	0.003	1
BE	gas	289.14	44.0	1.936	17	4.106	19	4.440	20
CZ	gas	288.88	29.0	1.947	7	3.947	17	3.947	17
CZ	hard_coal	288.15	36.0	1.941	9	3.748	9	3.748	9
CZ	lignite	289.3	38.0	1.919	18	3.772	25	4.494	25
DE	gas	284.6	17.0	11.287	28	17.077	18	16.972	18
DE	hard_coal	284.14	84.0	20.107	62	24.395	11	22.704	48
DE	lignite	1125.56	96.0	18.238	18	18.59	15	18.59	15
DE	other_fossil	1810.1	30.0	0.6	2	0.905	4	2.341	11
DK	gas	229.78	10.0	0.108	1	0.108	1	0.108	1
DK	hard_coal	171.84	63.0	2.151	6	2.895	8	2.123	7
DE	other_fossil	1027.87	108.0	1.971	15	2.251	15	2.251	15
ES	gas	280.63	48.0	21.466	62	24.671	72	24.671	72
ES	hard_coal	1046.87	88.0	7.622	18	9.191	16	6.191	16
ES	hard_coal	154.59	47.0	0.891	4	2.274	13	2.274	13
ES	other_fossil	759.61	14.0	0.13	1	0.13	1	0.441	10
ES	gas	286.38	48.0	6.881	16	6.12	18	6.211	18
ES	hard_coal	108.83	71.0	2.89	4	2.89	1	2.89	1
GB	gas	288.64	64.0	20.851	12	23.861	77	34.515	84
GB	hard_coal	1125.0	18.0	14.79	18	12.081	11	13.223	17
GB	gas	457.38	36.0	3.779	18	4.899	14	4.899	14
GB	lignite	1439.22	108.0	1.905	14	1.905	14	1.905	14
GR	gas	279.81	62.0	1.632	8	1.761	10	1.761	10
GR	lignite	1100.83	84.0	0.633	2	0.633	2	0.633	2
GR	gas	289.89	44.0	1.775	16	1.965	14	1.975	14
GR	hard_coal	1010.2	88.0	0.855	2	0.855	3	0.855	3
FR	gas	285.17	52.0	23.148	18	27.175	88	28.462	72
FR	hard_coal	887.47	15.0	1.081	15	4.926	18	6.536	18
FR	other_fossil	468.37	22.0	1.479	8	1.94	10	10.034	43
NL	gas	283.24	46.0	7.578	18	12.707	14	14.371	17
NL	hard_coal	1188.07	24.0	0.791	1	4.662	6	4.662	6
PL	gas	270.16	38.0	0.532	2	0.657	5	1.572	15
PL	hard_coal	942.41	16.0	12.054	12	14.901	16	18.861	32
PL	lignite	1108.45	78.0	1.576	18	7.064	18	8.476	24
PT	gas	461.13	61.0	2.838	7	1.828	10	1.828	10
RO	gas	454.42	28.0	1.108	9	1.428	21	1.41	21
RO	hard_coal	1127.46	71.0	0.76	4	1.193	6	1.193	6
RO	lignite	1095.76	86.0	1.108	13	4.011	14	4.011	14
SE	gas	461.8	44.0	0.576	1	0.576	1	0.576	1
SE	hard_coal	983.89	36.0	0.12	2	0.12	2	0.12	2
SE	lignite	1245.15	87.0	0.12	2	0.12	2	0.12	2

Technology	Subsumed ENTSO-E technologies
hard_coal	Fossil hard coal
lignite	Fossil brown coal/lignite
gas	Fossil gas
other_fossil	Fossil coal-derived gas, fossil peat, other, fossil oil, fossil oil shale
nuclear	Nuclear
biomass	Biomass
waste	Waste
other_renewable	Geothermal, marine, other renewable
hydro	Hydro pumped storage, hydro run-of-river and poundage, hydro water reservoir
solar	Solar
wind_onshore	Wind onshore
wind_offshore	Wind offshore

Task 3:

Trends of Electricity Generation (2019 - 2024)

1. Visualize the mix of generation technologies for the mentioned years
 - ▶ Choose an adequate temporal resolution (hourly, weekly, monthly, yearly) and type of plot to depict the major trends.
 - ▶ Plot the CO₂ intensity of electricity (kg CO₂/kWh) in the same graph.
2. What kind of trends can you observe regarding:
 - ▶ Share of renewables
 - ▶ Share of gas plants
 - ▶ Share of coal plants
 - ▶ CO₂ intensity

→ Conduct a linear regression vs. time for these objects
3. Predict the date when:
 - ▶ the share of renewables is > 85%
 - ▶ the share of gas plants is < 10%
 - ▶ all coal plants are phased-out
 - ▶ the electricity in the grid is net CO₂-neutral

Task 4: MLR Regression Model for the Day-Ahead-Price



1. Create a regression model for the day-ahead-price using data for the year 2024. Only use training and validation data for 2024
2. Compare at least 5 different regression models of different model complexity
 - ▶ You can test different features
 - ▶ You may engineer new features tapping other available resources/data sets
 - ▶ Use a systematic approach to identify those features of highest relevance
 - ▶ Avoid overfitting
 - ▶ Perform model evaluation using k-fold Cross-Validation
3. Evaluate best model
 - ▶ Select best model
 - ▶ Fit best model
 - ▶ Evaluate the goodness of fit
 - ▶ Perform model evaluation using k-fold Cross-Validation
 - ▶ Describe the final model, its structure and the reason why it was selected.

Tips on using dates and timestamps



- ▶ It might be useful to convert datetime (python class) to Unix timestamp (integer in seconds) and vice versa
- ▶ You can import the csv file and convert the timestamp automatically

```
from datetime import datetime

# Create a datetime object
dt = datetime(2024, 12, 16, 10, 30)
# Example datetime (year, month, day, hour, minute)

# Convert to Unix timestamp
unix_timestamp = dt.timestamp()

print("Unix Timestamp:", unix_timestamp)

# Convert Unix timestamp to datetime in UTC
unix_timestamp = 1702725000 # Replace with your timestamp
dt_utc = datetime.fromtimestamp(unix_timestamp, tz=datetime.timezone.utc)

print("Datetime in UTC:", dt_utc)

# Automatically parse dates from column 'timestamp'
df = pd.read_csv(csv_file, parse_dates=['timestamp'])
```